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Intelligent Data Processing and Applications

COE 543/743

Chapter 8 - Semantic Similarity Evaluation

Measures and Applications

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Computer Software Engineering

- ◆ **Introduction**
- ◆ **Corpus-based Semantic Analysis**
- ◆ **Knowledge-based Semantic Analysis**
- ◆ **Applications**
- ◆ **Conclusion**



◆ Information needs are increasing exponentially

- ◆ Digital information becomes pervasive with continuous exposure to the Web

Problem

The value of information depends on how easy it is to search and manage...

How to handle the huge amounts of information ?

Automatic extraction of Semantic Meaning becomes crucial

◆ Promised land: Semantic Web

- ◆ Extending the Web by assigning information well defined meaning
 - Render the Web machine-readable and easier to process
 - Effective data search and automated management

Overview

Introduction

Corpus-based

Knowledge -based

Applications

Conclusion

- ◆ **Semantic Web requires:**
 - ◆ Computation of semantic meaning
 - **Semantic reference sources**
 - **Methods of semantic similarity**

Overview

Introduction

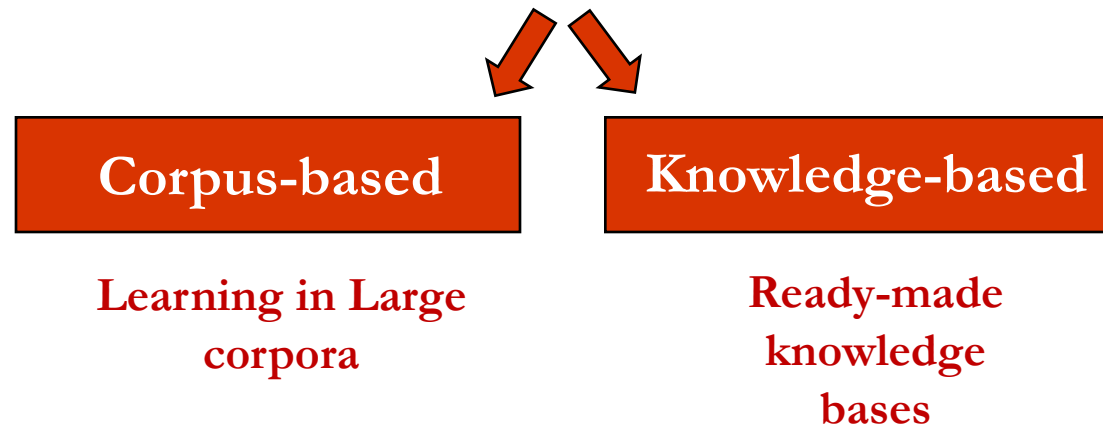
Corpus-based

Knowledge -based

Applications

Conclusion

Semantic Analysis



- ◆ Introduction
- ◆ **Corpus-based Semantic Analysis**
- ◆ Knowledge-based Semantic Analysis
- ◆ Applications
- ◆ Conclusion



- ◆ Semantics identified based on word distribution in text corpus

Overview

Introduction

Corpus-based

Knowledge
-based

Applications

Conclusion

Main Premise

Semantically similar words have similar syntactic distributions in a corpus



- Co-occurrence
- Contextual
- Lexical patterns

- Decision trees
- Bayesian networks
- Support vector machines

- Variance
- Correlation
- Regression analysis

Overview

Introduction

Corpus-based

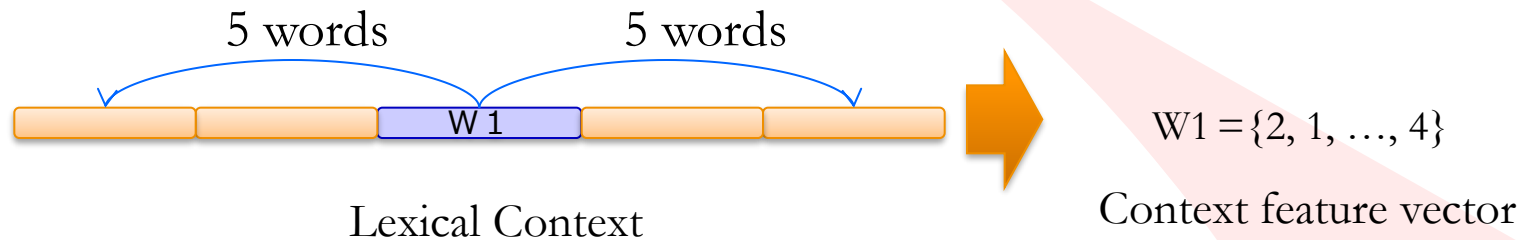
Knowledge
-based

Applications

Conclusion

◆ Main idea: **context-based** similarity

- ◆ **Step 1:** Syntactic measures define individual word descriptions



- ◆ **Step 2:** Statistical analysis normalizes word descriptions over corpus
- ◆ **Step 3:** Learning considers human perception of semantics
 - Limiting the effects of lexico-syntactic and stylistic dependency of words
 - To go beyond the surface of appearance of words

Overview

Introduction

Corpus-based

Knowledge
-based

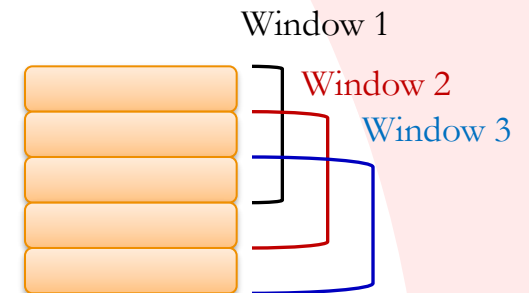
Applications

Conclusion

◆ Main idea: co-occurrence similarity

- ◆ **Step 1:** Create textual windows of a certain number of words or sentences
- ◆ **Step 2:** Count number of occurrence of every target words in textual windows
- ◆ Step 3: Count number of co-occurrence of target words in textual windows
 - ▢ Underlining how often the words occur together

$$\text{sim}(w_1, w_2) = \log \frac{\text{Count}(w_1, w_2)}{\text{Count}(w_1) \times \text{Count}(w_2)}$$



Overview

Introduction

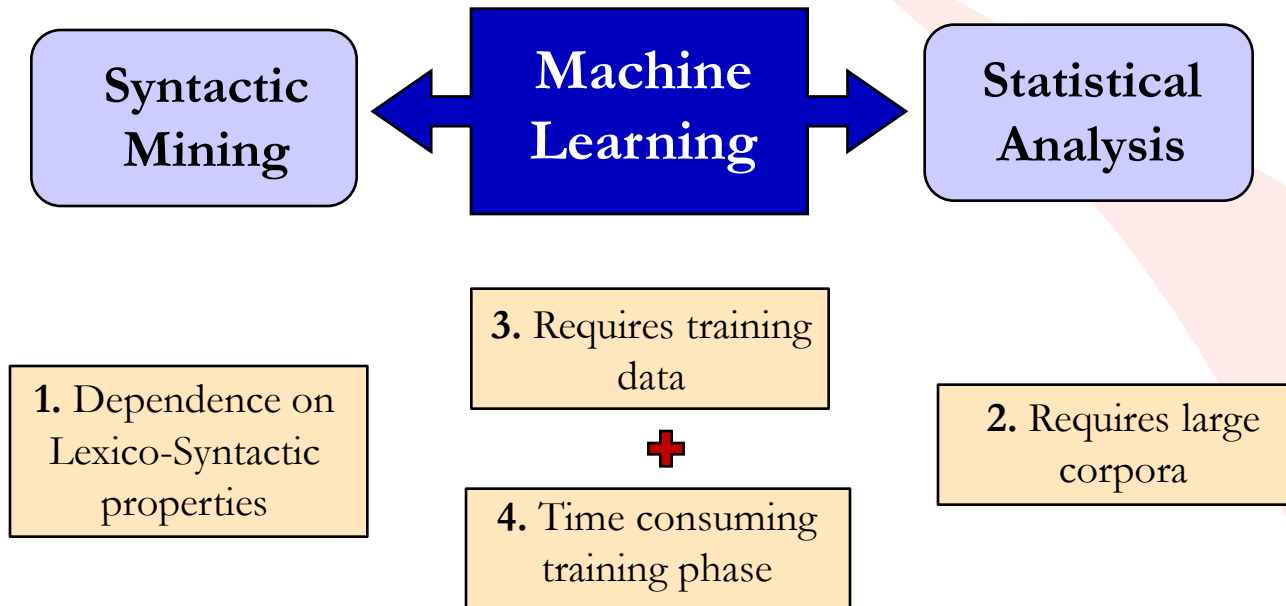
Corpus-based

Knowledge
-based

Applications

Conclusion

- ◆ Corpus-based methods have been popular
 - ◆ Yet have suffered from various limitations:



- ◆ Improvements using:
 - ◆ Labeled corpora
 - ◆ Bootstrapping



Country	Country	Country	Country	Country	Country	Org	Date
Iraq	Morocco	Oman	Saudi Arabia	the Sudan	Yemen	ESCWA	2014

Non-communicable diseases are rising at alarming rates, and the number of people with dementia is increasing. The health system and care delivery services are under pressure as infectious diseases persist, mainly in the least developed countries.

1. (7) **poverty**, poorness, impoverishment -- (the state of having little or no money and few or no material possessions)

In a context of prevailing and increasing poverty in many Arab countries, social protection schemes, present serious weaknesses.

lack universal coverage, most social security schemes are linked to formal employment and are contributory; and modest; are contributory; and widespread informal sector employment is not covered. Furthermore, their financial sustainability is in question in view of the foreseen growing ageing population.

1. (24) failing, **weakness** -- (a flaw or weak point; "he was quick to point out his wife's failings")
2. (2) helplessness, **weakness**, impuissance -- (powerlessness revealed by an inability to act; "in spite of their weakness the group remains highly active")
3. (2) **weakness** -- (the property of lacking physical or mental strength; liability to failure under pressure or stress or strain; "his weakness increased as he became older"; "the weakness of the span was overlooked until it collapsed")
4. **weakness** -- (the condition of being financially weak; "the weakness of the dollar against the yen")
5. **weakness** -- (a penchant for something even though it might not be good for you; "he has a weakness for chocolate")

Table 2 sets out data on the proportion of older women and men, above statutory pensionable age, receiving an old age pension. It clearly shows

- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ **Knowledge-based Semantic Analysis**
- ◆ Applications
- ◆ Conclusion



Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Main premise

Predefined semantic knowledge is required to identify semantic meaning

- ◆ Relies on predefined structured sense inventories

◆ Building blocks:

- ◆ Machine-readable lexical knowledge bases
 - Ready-made sources of information about word senses
- ◆ Semantic similarity measures in knowledge bases
 - Evaluating relatedness between word senses

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

◆ Knowledge base

◆ Controlled vocabulary

- A list of ordered words with explicit semantic meanings
 - Machine-readable dictionary

◆ Taxonomy

- A controlled vocabulary organized in a structural hierarchy
 - Using specialization (*Is-A*) links and containment (*Part-Of*) links

◆ Thesaurus

- A taxonomy enriched with non-hierarchical relations
 - Associative relations: Related-To, Attribute-Of, See-Also, etc.

◆ Ontology

- Taxonomy/Thesaurus with an explicit formalization of relations
 - Using dedicated grammar rules

Overview

Introduction

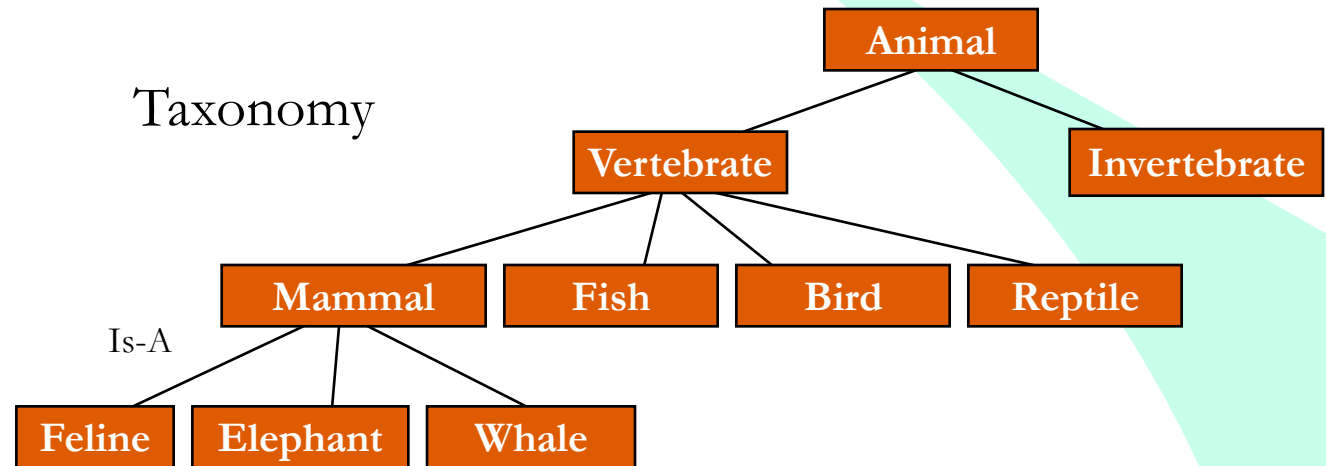
Corpus-based

Knowledge-based

Applications

Conclusion

Taxonomy



Overview

Introduction

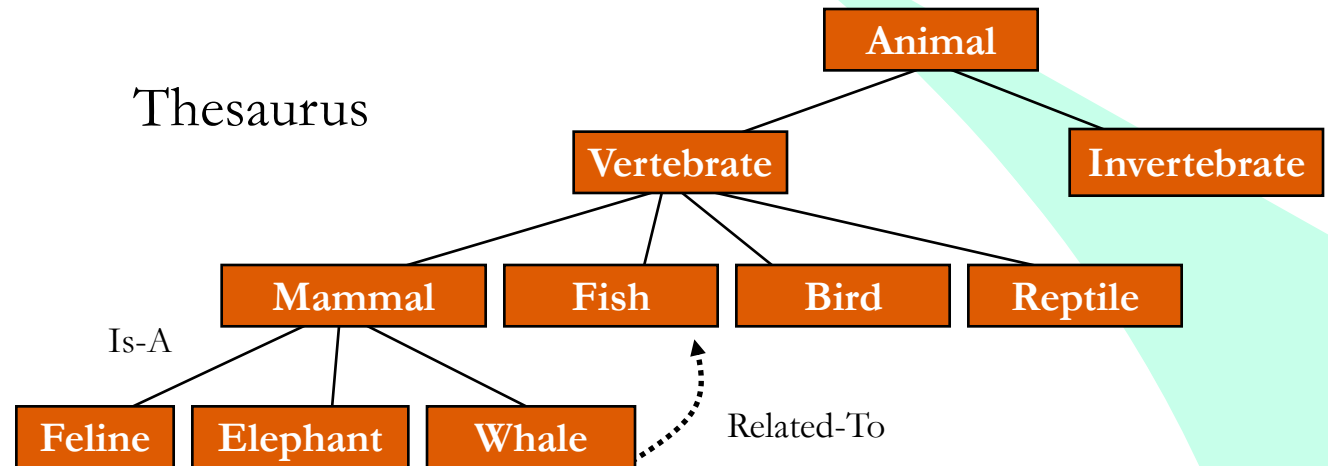
Corpus-based

Knowledge-based

Applications

Conclusion

Thesaurus



Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Challenge

How to exploit knowledge bases in computing semantic meaning ?



Semantic Concept Matching

- ◆ **Semantic similarity measures**
 - ◆ Automatically identify semantically related concepts
 - ◆ Rank semantically related concepts following their degrees of similarity

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

◆ Main approach

- ◆ **Hierarchical knowledge base (Taxonomy)**
 - Modeled as a Semantic Network
 - Concepts connected via (hierarchical) relations
- ◆ **Each concept underlines a set of synonymous words**
 - Words having the same semantic meaning
 - Each concept is associated a textual definition: Gloss

Overview

Introduction

Corpus-based

Knowledge-based

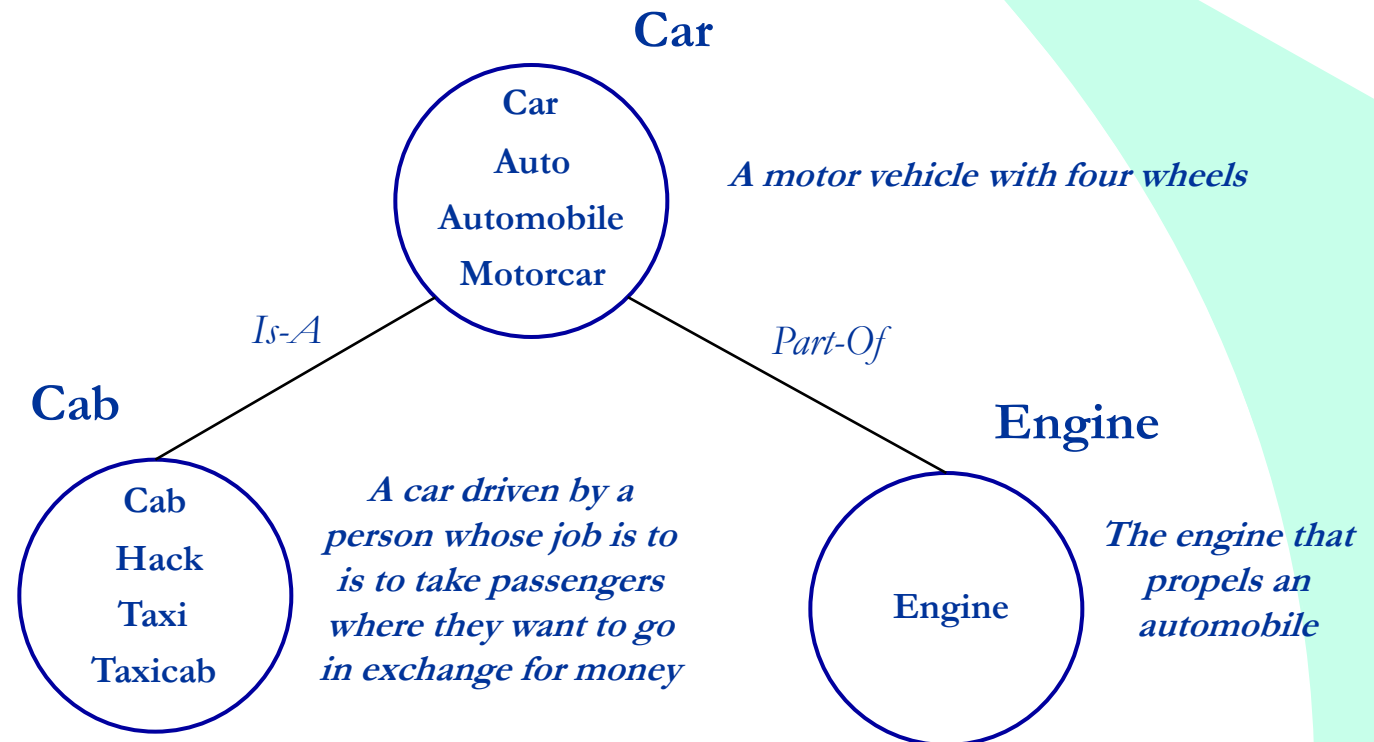
Applications

Conclusion

◆ Main approach

◆ Hierarchical knowledge base (Taxonomy)

- Modeled as a Semantic Network
 - Concepts connected via (hierarchical) relations



Overview

Introduction

Corpus-based

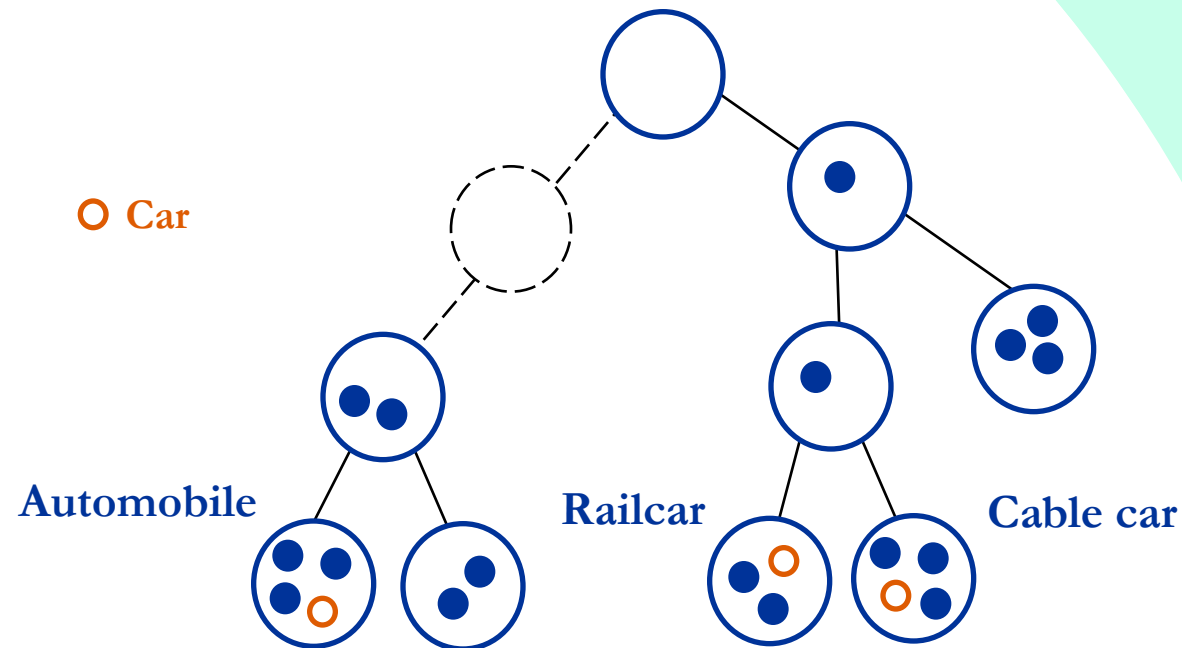
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Applications

Conclusion

◆ Main Approach

- ◆ A word may correspond to different concepts
 - Multiple senses
 - Homonymy (Polysemy)



Overview

Introduction

Corpus-based

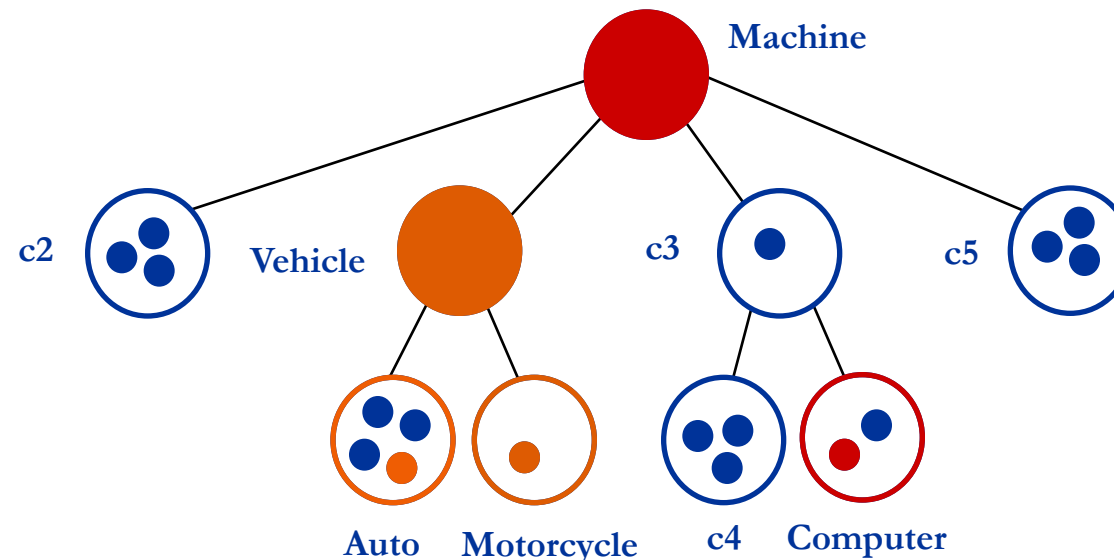
Knowledge-based

Applications

Conclusion

◆ Main Approach

- ◆ The semantic similarity between words
 - **Similarity between corresponding concepts in the semantic network**
- ◆ Semantic similarity between concepts depends on the amount of information they share in common
 - Taxonomy: the **Lowest Common Ancestor**



- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ **Knowledge-based Semantic Analysis**
- ◆ Applications
- ◆ Conclusion

- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ Knowledge-based Semantic Analysis
 - ◆ **Edge-based similarity**
 - ◆ Node-based similarity
 - ◆ Gloss-based similarity
- ◆ Applications
- ◆ Conclusion

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Edge-based Approach

◆ Premise

- ◆ Semantic similarity between two concepts c_1 and c_2
 - ▢ Minimum distance (in number of links) in semantic network
 - Identifying, in a roundabout way, the **Lowest Common Ancestor**

◆ Basic mathematical formula:

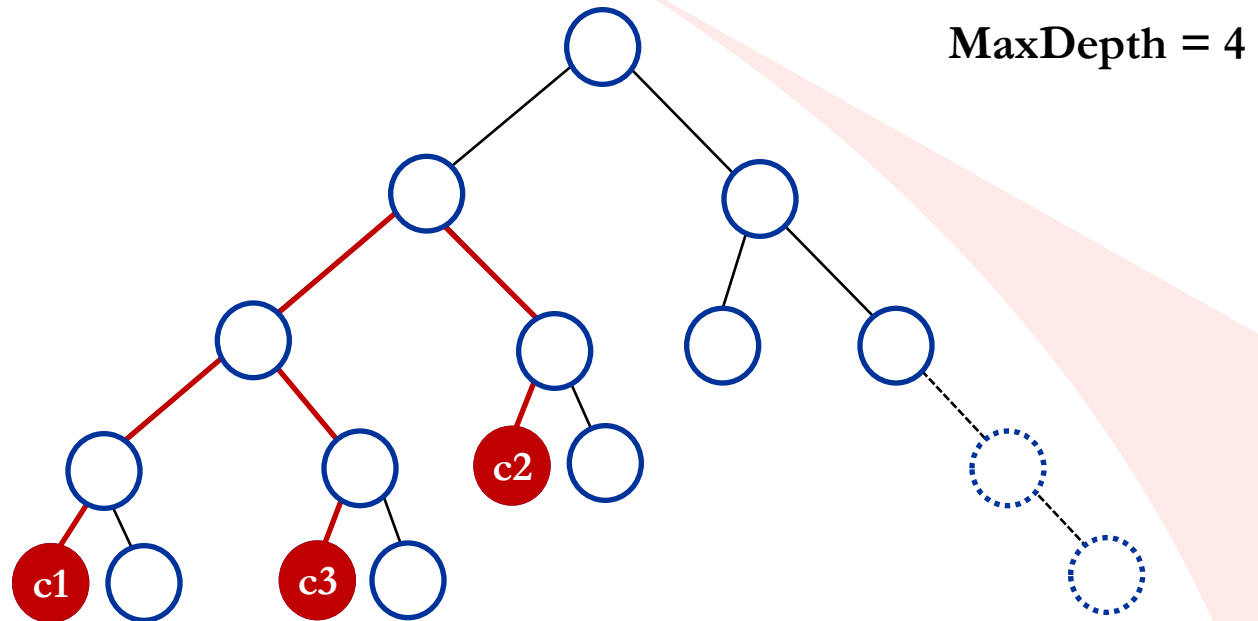
$$\text{Sim}_{\text{Edge}}(c_1, c_2) = 1 - \frac{\text{MinPath}(c_1, c_2, \text{SN})}{2 \times \text{MaxDepth}(\text{SN})}$$

Minimizing Distance



Maximizes Similarity

◆ Example



$$\text{◆ } \text{Sim}_{\text{Edge}}(c_1, c_2) = 1 - \frac{\text{MinPath}(c_1, c_2)}{2 \times \text{MaxDepth}} = 1 - \frac{5}{8} = 0.375$$

$$\text{◆ } \text{Sim}_{\text{Edge}}(c_1, c_3) = 1 - \frac{\text{MinPath}(c_1, c_3)}{2 \times \text{MaxDepth}} = 1 - \frac{4}{8} = 0.5$$

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Overview

Introduction

Corpus-based

Knowledge-based

Applications

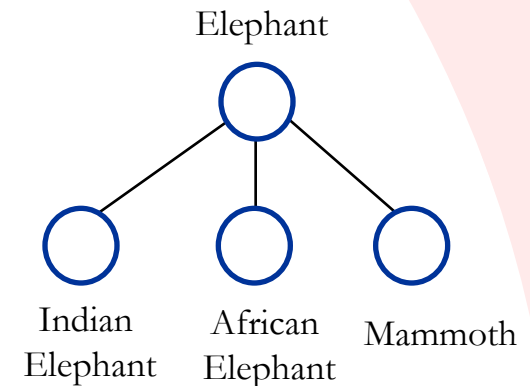
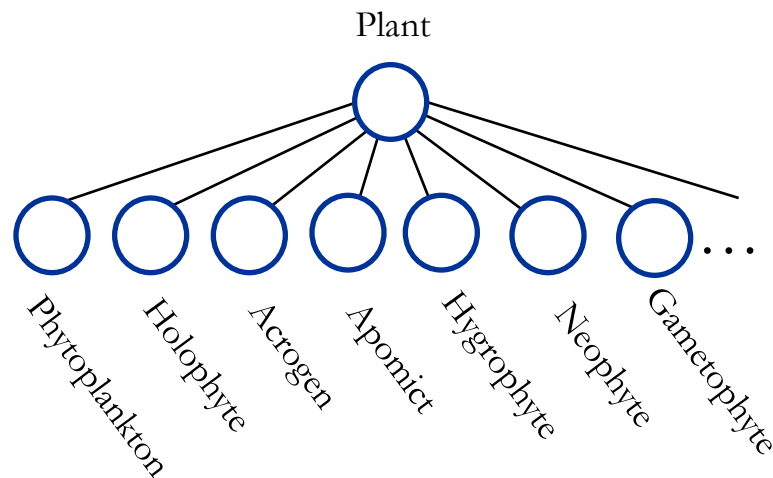
Conclusion

♦ Advantage

- ♦ Simple and intuitive approach

♦ Main limitation

- ♦ Links underline uniform distances
 - Which is not always semantically accurate
- ♦ In real semantic networks, link distance may vary following:
 - ♦ Network density



Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

◆ Advantage

- ◆ Simple and intuitive approach

◆ Main limitation

- ◆ Links underline uniform distances
 - Which is not always semantically accurate

◆ In real semantic networks, link distance may vary following:

- ◆ Network density
- ◆ Network depth

Overview

Introduction

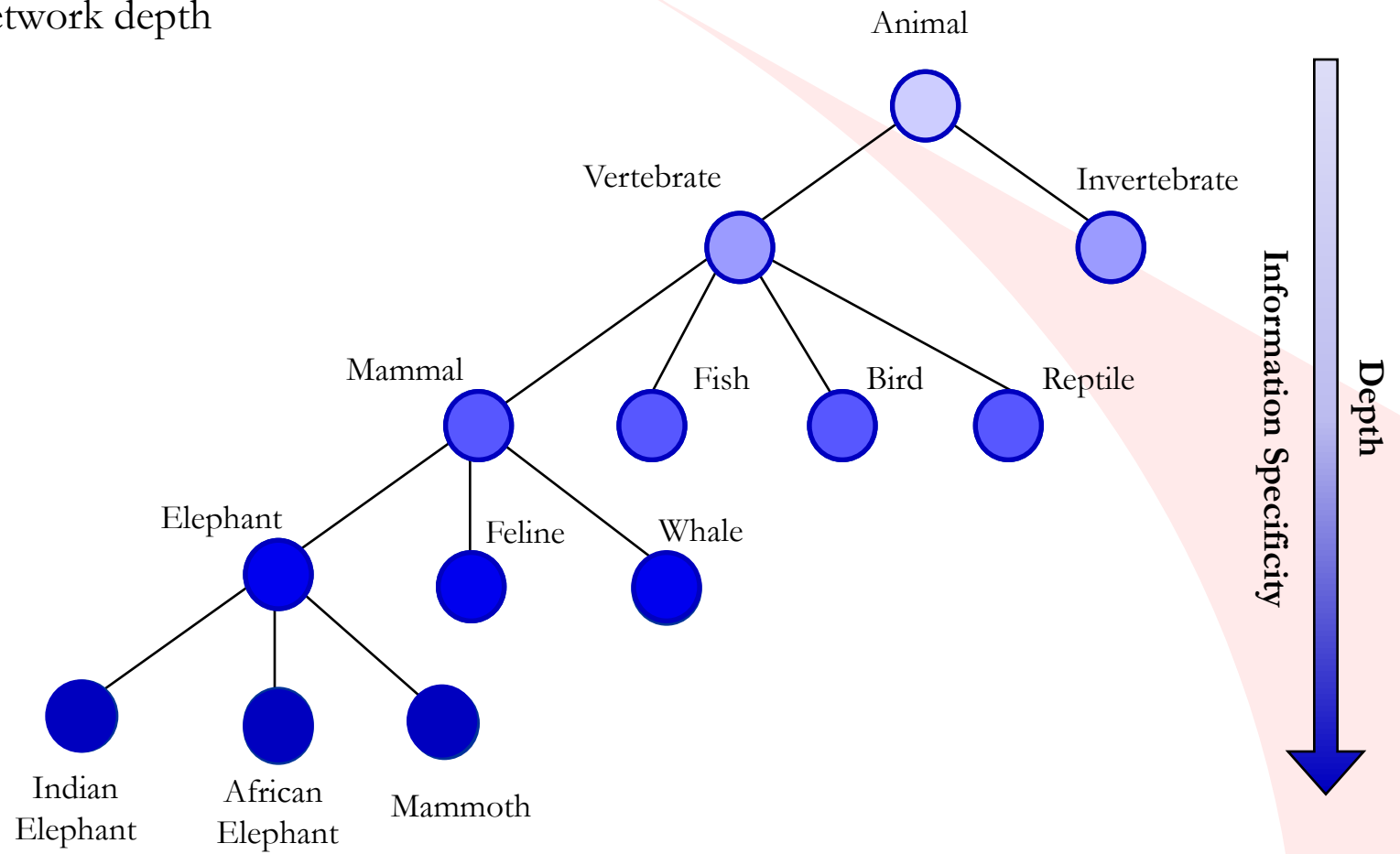
Corpus-based

Knowledge-based

Applications

Conclusion

◆ Network depth



Overview

Introduction

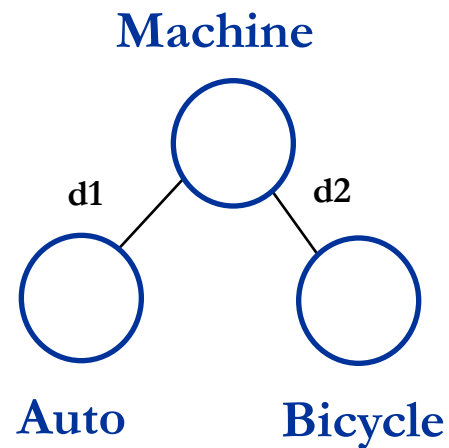
Corpus-based

**Knowledge-
based**

Applications

Conclusion

- ◆ In real semantic networks, link distance may vary following:
 - ◆ Network density
 - ◆ Network depth
 - ◆ User perception of semantic meaning



*Automobiles are more
mechanized than bicycles*

Seems more intuitive
to have $d1 < d2$

Overview

Introduction

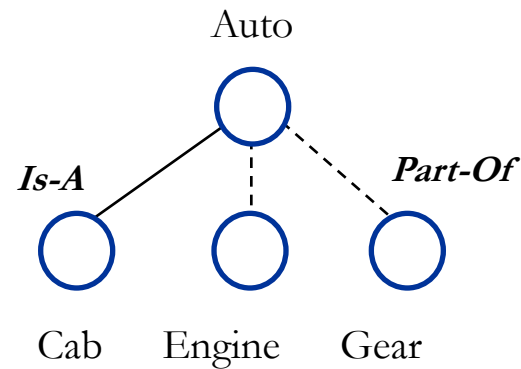
Corpus-based

**Knowledge-
based**

Applications

Conclusion

- ◆ In real semantic networks, link distance may vary following:
 - ◆ Network density
 - ◆ Network depth
 - ◆ User perception of semantic meaning
 - ◆ Link Type



Overview

Introduction

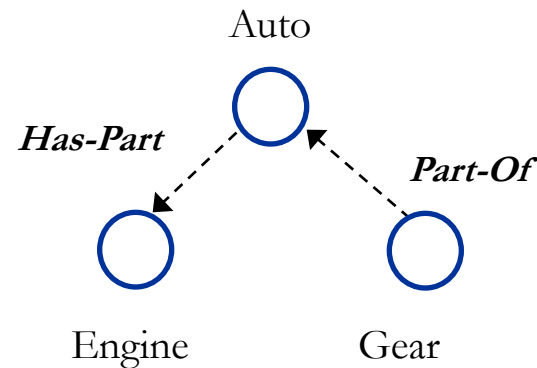
Corpus-based

**Knowledge-
based**

Applications

Conclusion

- ◆ In real semantic networks, link distance may vary following:
 - ◆ Network density
 - ◆ Network depth
 - ◆ User perception of semantic meaning
 - ◆ Link Type
 - ◆ Link direction



Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Edge-based Approach

◆ Some approaches have suggested weighting links

- ◆ To consider varying semantic distances

◆ Sample mathematical formula:

$$\text{Dist}(c_1, c_2, \text{SN}) = \sum_{c \in \text{MinPath}(c_1, c_2, \text{SN})} \text{Weight}(c_1, c, \text{SN})$$

$$\text{Weight}(c_1, c_2, \text{SN}) = \left(\beta + (1-\beta) \times \frac{\text{Child}(\text{SN})}{|\text{Child}(c_2)|} \right) \times \left(\frac{\text{Depth}(c_2) + 1}{\text{Depth}(c_2)} \right)^\alpha \times \text{Type}(c_1, c_2)$$

How to weight different linking factors ?

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Edge-based Approach

◆ Advantage

- ◆ Simple and intuitive

◆ Limitations

- ◆ Uniform link distances *versus* Varying link distances
 - Concept Depth
 - Network Density
 - Link Type
 - Link Direction
 - User perception
- ◆ How to combine different link weighting factors

- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ Knowledge-based Semantic Analysis
 - ◆ Edge-based similarity
 - ◆ **Node-based similarity**
 - ◆ Gloss-based similarity
- ◆ Applications
- ◆ Conclusion

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Node-based Approach

◆ Premise

- ◆ Augment information in Semantic Network
 - Integrating corpus-based statistics

Get round the problem of varying link distances

◆ Similarity is evaluated based on **Information Content**

- ◆ Function of the occurrence probability of concepts in a text corpus
 - $IC(c1) = -\log(p(c1))$

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

◆ Main idea:

- ◆ Augment the taxonomy with a probability function P

- $P : \text{Taxonomy} \rightarrow [0, 1]$
- $p(c)$ = occurrence probability of concept c

- $$p(c) = \frac{\text{freq}(c)}{N}$$

- $\text{freq}(c) = \sum_{\text{For all words } w \in \text{concept } c} \text{Count}(w)$

- N = total number of words in text corpus

- ◆ Similarity between concepts c_1 and c_2

- $$\text{Sim}_{\text{Node}}(c_1, c_2) = \max_{c \in S(c_1, c_2)} [-\log p(c)]$$

- with $S(c_1, c_2)$ = set of common ancestors of c_1 et c_2

Overview

Introduction

Corpus-based

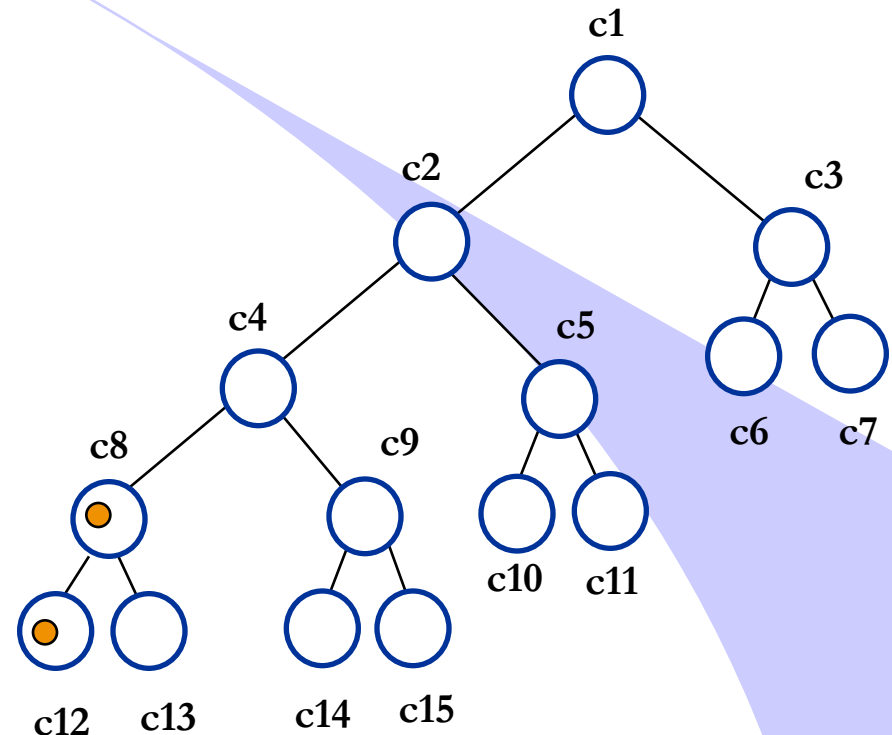
Knowledge-based

Applications

Conclusion

◆ Example

- ◆ $\text{freq}(c_{12}) = 0 + 1$
- ◆ $\text{freq}(c_8) = 0 + 1 + 1$
- ◆ $\text{freq}(c_4) = 0 + 1 + 1$
- ◆ $\text{freq}(c_2) = 0 + 1 + 1$
- ◆ $\text{freq}(c_1) = 0 + 1 + 1$
- ◆ ...
- ◆ $N = 0 + 1 + 1$



- ◆ For each concept c_1 ancestor of c_2 , $\text{freq}(c_1) \geq \text{freq}(c_2)$
 - ◆ $p(c_1) \geq p(c_2)$
 - $-\log(p(c_1)) \leq -\log(p(c_2))$

Ancestors are less informative than descendants

Overview

Introduction

Corpus-based

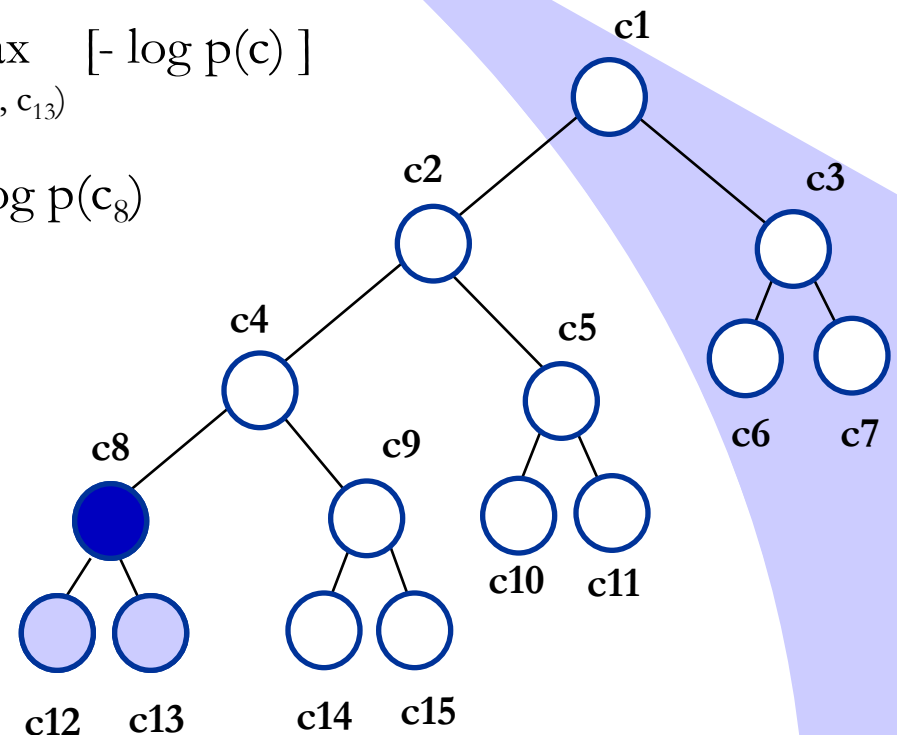
Knowledge-based

Applications

Conclusion

LCA maximizes Information Content

- ◆ Similarity between concepts c_{12} and c_{13}
 - $\text{Sim}_{\text{Node}}(c_{12}, c_{13}) = \max_{c \in S(c_{12}, c_{13})} [-\log p(c)]$
 $= -\log p(c_8)$



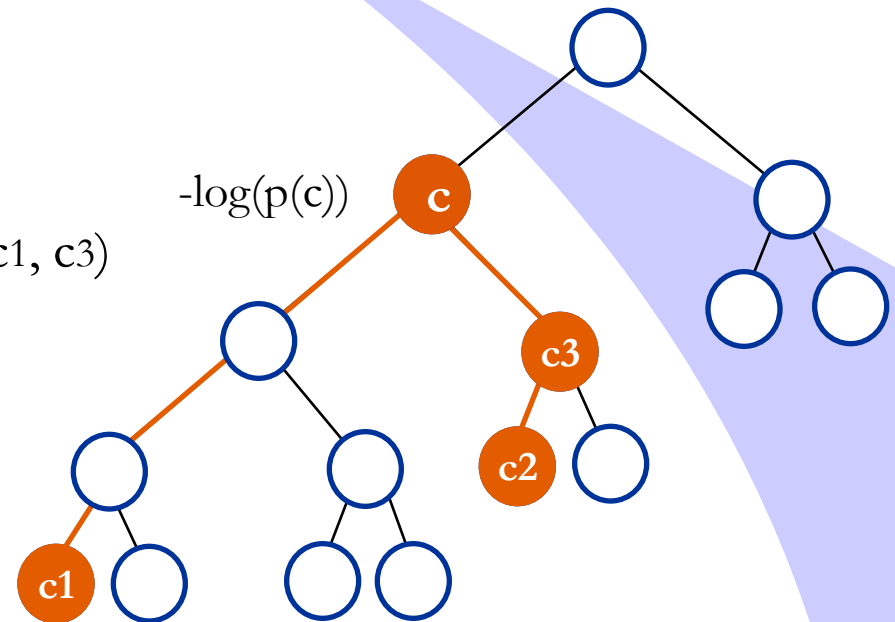
Overview

Corpus-based

Applications

Conclusion

- $$\text{Sim}_{\text{Note}}(c_1, c_2) \stackrel{?}{=} \text{Sim}_{\text{Node}}(c_1, c_3)$$



- 38-

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Improved measure

◆ Follows basic assumptions:

- ◆ Similarity between concepts depends on their commonality
- ◆ Similarity between concepts depends on their difference
- ◆ Similarity is maximum when concepts are identical

◆ Mathematical formula [1]:

$$\text{◆ } \text{Sim}_{\text{Node+}}(c1, c2) = \frac{2 \log P(c)}{\log P(c1) + \log P(c2)}$$

□ c is lowest common ancestor of c1 and c2

Overview

Introduction

Corpus-based

Knowledge-based

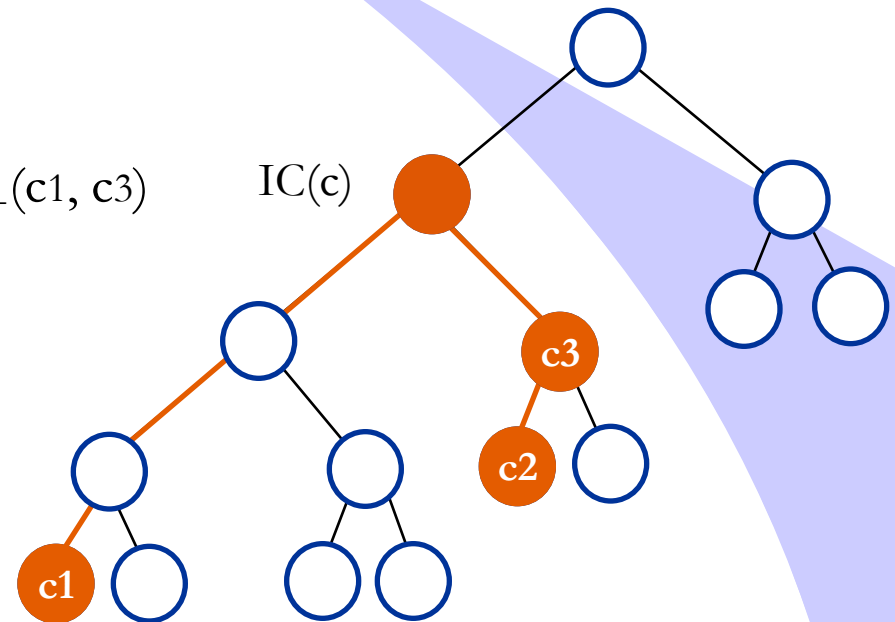
Applications

Conclusion

◆ Improved measure

- ◆ Considers information content of LCA and own concepts

$$\text{Sim}_{\text{Node}+}(c1, c2) \leq \text{Sim}_{\text{Node}+}(c1, c3)$$



- ◆ Empirical results show improved results in comparison with
 - ◆ Basic node-based measure
 - ◆ Edge-based measures

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Node-based Approach

◆ Advantages

- ◆ Getting round the varying link distance problem
- ◆ Integrating corpus-based statistics
 - Adapting the knowledge based to specific corpora
- ◆ Empirical results show improved quality in comparison with edge-based methods

◆ Limitations

- ◆ Requires an input text corpus
- ◆ Requires additional computation time to compute corpus-statistics

Corpus-statistics can be computed offline

Overview



- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ Knowledge-based Semantic Analysis
 - ◆ Edge-based similarity
 - ◆ Node-based similarity
 - ◆ **Gloss-based similarity**
- ◆ Applications
- ◆ Conclusion

Overview

Introduction

Corpus-based

**Knowledge-
based**

Applications

Conclusion

Gloss-based Approach

◆ Premise

- ◆ Evaluate the word overlap between the glosses of concepts
 - A gloss underlining the textual definition describing a concept
 - E.g., the gloss of the 1st sense of word car in WordNet is “*A motor vehicle with four wheels*” (cf. slide 15)

◆ Basic mathematical formula: **original method**

- ◆ $\text{Sim}_{\text{Gloss}}(c_1, c_2) = |\text{gloss}(c_1) \cap \text{gloss}(c_2)|$
 - Other set-based or vector-based similarity measures can be used
 - Jaccard, Dice, Cosine, etc. (cf. Chapter 4)

Gloss-based Approach

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

- ◆ **Problem:** The original gloss-based measure compares glosses by counting the number of words that are shared between them
 - ◆ It does not differentiate between single word and phrasal overlaps
 - Treating each gloss as a *bag of words*
- ◆ **Improvement:** based on Zipfian [1] relationship between phrases
 - ◆ *The longer the phrase, the less likely it is to occur multiple times in a given corpus*
 - An n -word overlap is much rarer than a single word overlap
 - Assigning an n -word overlap the score of n^2
 - Giving it a score higher than the sum of the scores assigned to those n words when they occur separately

Overview

Introduction

Corpus-based

**Knowledge-
based**

Applications

Conclusion

Gloss-based Approach

◆ Improved method:

- ◆ $\text{Sim}_{\text{Gloss+}}(c_1, c_2) = |\text{gloss}(c_1) \cap_{\text{LCS}} \text{gloss}(c_2)|$
- ◆ **Overlap operator** defined based on Longest Common Sequence
 - Of one or more consecutive words that occurs in both glosses
- ◆ Overall process: given two strings:
 1. The **LCS** between them is **detected** and **removed**
 - A unique marker is placed in its place in each of the two input strings
 2. The two strings thus obtained are then again checked for overlaps
 - The process is **repeated** until there are no longer any overlaps
 3. The sizes of the overlaps thus found are squared and added together to compute the similarity score

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

Gloss-based Approach

◆ **Example:** Considering concepts *drawing paper* and *decal*

- ◆ $\text{gloss}(\text{drawing paper}) = \text{paper that is specially prepared for use in drafting}$
- ◆ $\text{gloss}(\text{decal}) = \text{art of transferring designs from specially prepared paper to a wood or glass or metal surface.}$

◆ Using original measure:

- ◆ $\text{Sim}_{\text{Gloss}}(\text{drawing paper}, \text{decal}) = 3$
 $\Rightarrow$ There are three words that overlap: *paper*, *specially* and *prepared*

◆ Using improved measure:

- ◆ $\text{Sim}_{\text{Gloss}+}(\text{drawing paper}, \text{decal}) = 5 = 4 + 1$
 - Score = 2^2 for the the overlap of the 2-word phrase *specially prepared*
 - Score = 1 for the the overlap of single word *paper*

Gloss-based Approach

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

◆ **Limitation** with both **original** and **improved methods**:

- ◆ They do not consider semantic relationships between concepts

◆ **Extended method:**

- ◆ $\text{Sim}_{\text{Gloss++}}(c_1, c_2) = |\text{ext_gloss}(c_1) \cap_{\text{LCS}} \text{ext_gloss}(c_2)|$ where:
 - $\text{ext_gloss}(c) = \text{gloss}(c) \cup \text{syn}(c) \cup \text{gloss}(\text{rel}(c))$
 - $\text{Syn}(c)$ is concept c 's set of synonyms
 - $\text{Rel}(c)$ is the set of concepts collected with c using semantic relationship rel
 - Rel can stand for *hypernymy*, *hyponymy*, *meronymy*, etc.

Gloss-based Approach

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

◆ Advantages:

- ◆ Considering **concept glosses** which were dismissed with edge and node-based measures
- ◆ Empirical results show **improved quality** in comparison with edge-based and, sometimes, node-based methods
 - Namely when applied in: **word sense disambiguation (WSD)**

◆ Limitations:

- ◆ Original/improved methods **do not consider semantic network structure**
- ◆ In the extended method: it is not clear **how to choose which semantic relations** to utilize in producing extended glosses

- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ Knowledge-based Semantic Analysis
 - ◆ Edge-based similarity
 - ◆ Node-based similarity
 - ◆ Gloss-based similarity
 - ◆ **Mathematical properties**
- ◆ Applications
- ◆ Conclusion

Overview

Introduction

Corpus-based

**Knowledge-
based**

Applications

Conclusion

◆ Formal definition of similarity

- ◆ Inverse of a metric distance function

◆ Metric properties

- ◆ $\text{Sim}(c_1, c_2) \leq \text{Sim}(c_1, c_1) \Rightarrow$ **Minimality**
- ◆ $\text{Sim}(c_1, c_2) = \text{Sim}(c_2, c_1) \Rightarrow$ **Symetricity**
- ◆ $\text{Sim}(c_1, c_3) \geq \text{Sim}(c_1, c_2) \times \text{Sim}(c_2, c_3) \Rightarrow$ **Triangular inequality**
 - ◆ $\text{Dist}(c_1, c_3) \leq \text{Dist}(c_1, c_2) + \text{Dist}(c_2, c_3)$

◆ Generalized Metric

- ◆ A distance (similarity) function that does not satisfy all metric axioms

Overview

Introduction

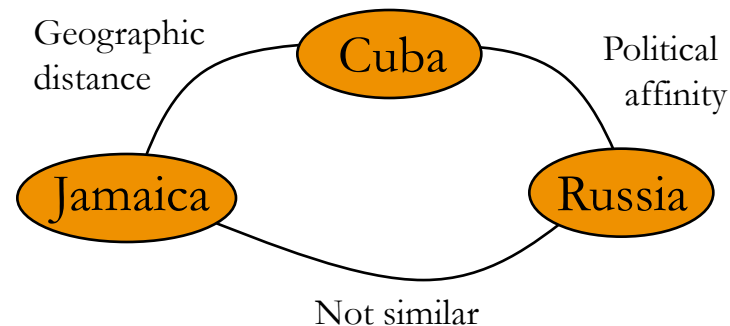
Corpus-based

Knowledge-based

Applications

Conclusion

- ◆ Most semantic similarity measures are generalized metrics
 - ◆ Semi-metrics
 - ◆ Do not satisfy **triangular inequality**
- ◆ That is due to the nature of semantic information
 - ◆ Rather than the similarity measures
 - ◆ Famous example by Tversky [1]:



- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ Knowledge-based Semantic Analysis
- ◆ Applications
- ◆ Conclusion

Semantic Web Vision

Overview

Introduction

Corpus-based

Knowledge-based

Applications

Conclusion

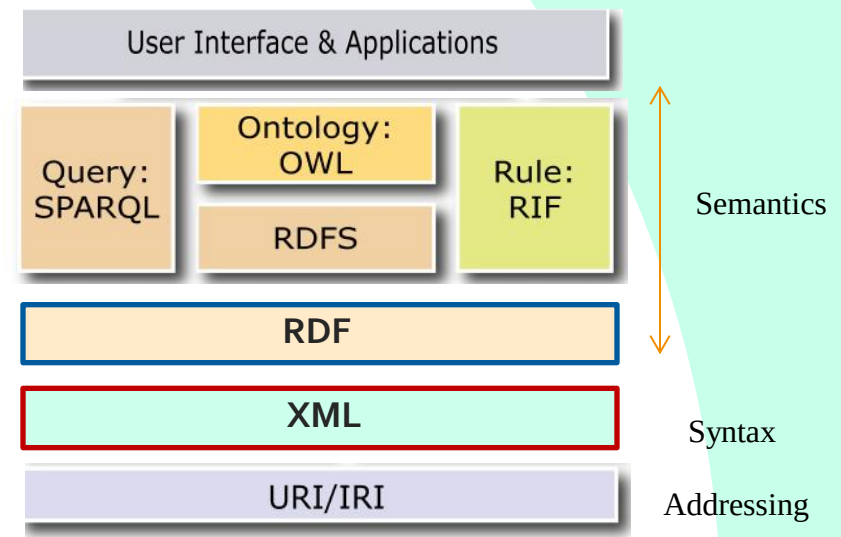
◆ Introduced by Tim Berners-Lee:

- Extending the Web by associating data with well-defined semantics
⇒ Better enabling computers and people to interact and work in cooperation

◆ Building blocks

SW standards to describe data and semantics:

- XML: Extensible Markup Language
- RDF: Resource Description Framework
- RDFS: RDF Schema
- OWL: Ontology Web Language
- SPARQL: RDF/OWL query language



Semantic Web Vision

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

- ◆ XML/JSON add **structure**, yet they **lack** formal **semantics**

```
<Department DeptId="15" DeptName="Sales">
  <Employee>
    <EmpNo>10</EmpNo>
    <FirstName>CHRISTINE</FirstName>
    <LastName>SMITH</LastName>
    <Phone>408-463-4963</Phone>
    <Salary>52750.00</Salary>
  </Employee>
  <Employee>
    <EmpNo>27</EmpNo>
    <FirstName>MICHAEL</FisrtName>
    <LastName>THOMPSON</LastName>
    <Phone>408-463-1234</Phone>
    <Salary>41250.00</Salary>
  </Employee>
</Department>
```

Semantic Web Vision

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

- ◆ **SW standards** are dedicated to **represent** formal **semantics** $\Rightarrow$ **knowledge**
 - ◆ With **RDF** for instance, **resources** are represented by **statements**
 - ◆ In the form of **subject-predicate-object** expressions: **triples**

Sample RDF statements extracted from
the Document tree in the previous slide

Subject	Predicate	Object
DeptId	Has	DeptName
DeptId	HasDept	15
Sales	IsA	DeptName
Sales	HasDeptId	15
Emp	Has	DeptId
Emp	HasEmpNo	27
Emp	HasEmpNo	10
...	...	...

Semantic Web Vision

Overview

Introduction

Corpus-based

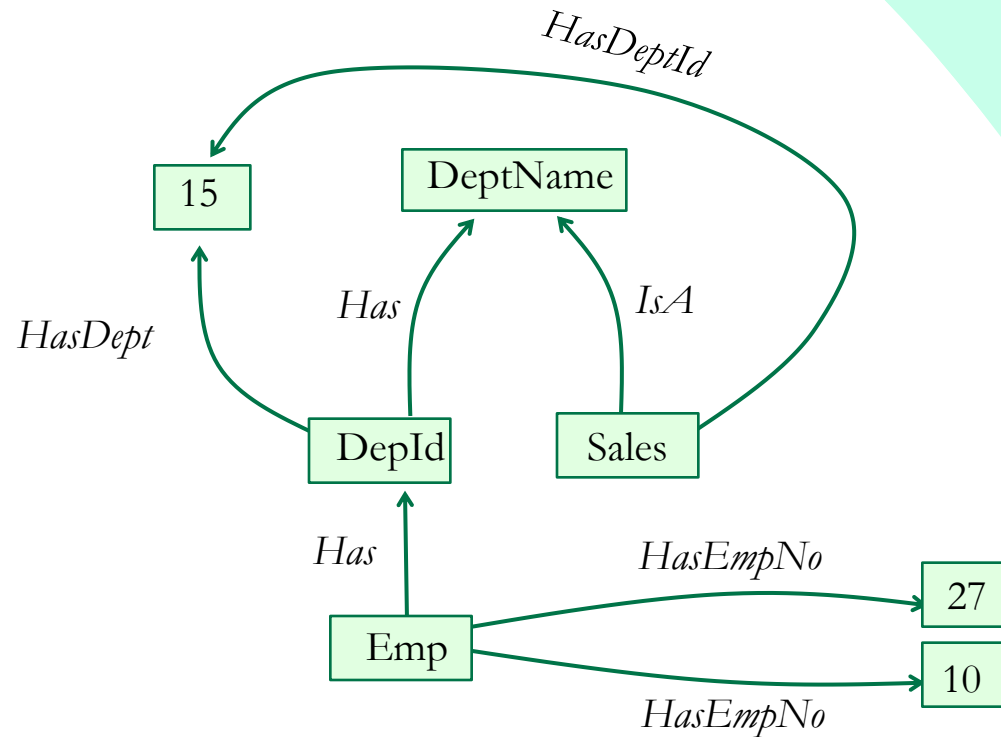
Knowledge-
based

Applications

Conclusion

◆ Generating and RDF graph

⇒ A Semantic Network



Semantic Web Vision

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

- ◆ **Data (information)** is usually coded in **semi-structured** syntax

- ◆ **Knowledge** is usually represented in **SW standards**

 - ◆ RDF, RDFS, and OWL



Also represented based on semi-structured document syntax

- ◆ **NOTE:** There is a fine line between:

 - ◆ Coding $\Rightarrow$ syntax (tagging, syntactic properties, etc.)
 - ◆ Representation model $\Rightarrow$ formalism (tree-based, graph-based, network-based, etc.)

- ◆ **Semi-structured tree-based** model is good for **data (information)**, yet it is not expressive enough for **knowledge!**

 - $\Rightarrow$ Hence the introduction of RDF and OWL graph-based models

Semantic Web Applications

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

◆ Information retrieval

- ◆ Lack of explicit semantics in state-of-the-art search engines
 - Semantic document indexing
 - Semantic query analysis
 - Pruning documents which may be syntactically similar , yet underline non-relevant meanings
 - **Precision**
 - Retrieving documents which may be syntactically different, yet share similar semantics
 - **Recall**

Semantic Web Applications

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

◆ Information extraction

- ◆ Extracting semantically related concepts from a corpus
 - Part-of-Speech tagging
 - Text categorization

◆ Machine translation

- ◆ The automatic identification of the correct translation of a word
 - Requires semantic evaluation
 - Helps the system choose better candidates

Live speech translation techniques

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

Semantic Web Applications

◆ Content analysis

- ◆ Classification of blogs
 - Identifying relevant topics
 - Organizing blogs following semantic categories
 - Identifying those most interesting to user requests
- ◆ Social network analysis
 - Discovering semantic connections
 - Entity disambiguation

◆ Lexicography

- ◆ Creating domain specific dictionaries



Mutual benefit

- ◆ Introduction
- ◆ Corpus-based Semantic Analysis
- ◆ Knowledge-based Semantic Analysis
- ◆ Applications
- ◆ Conclusion

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

◆ Web information is becoming pervasive

We are drowning in data, yet we starve of information!

The value of information depends on how easy it is to search and manage...

◆ **Promised land:** Semantic Web

- ◆ Extending the Web by assigning information well defined meaning
 - Effective data search and automated management

Computation of semantic meaning requires

Semantic similarity evaluation

Overview

Introduction

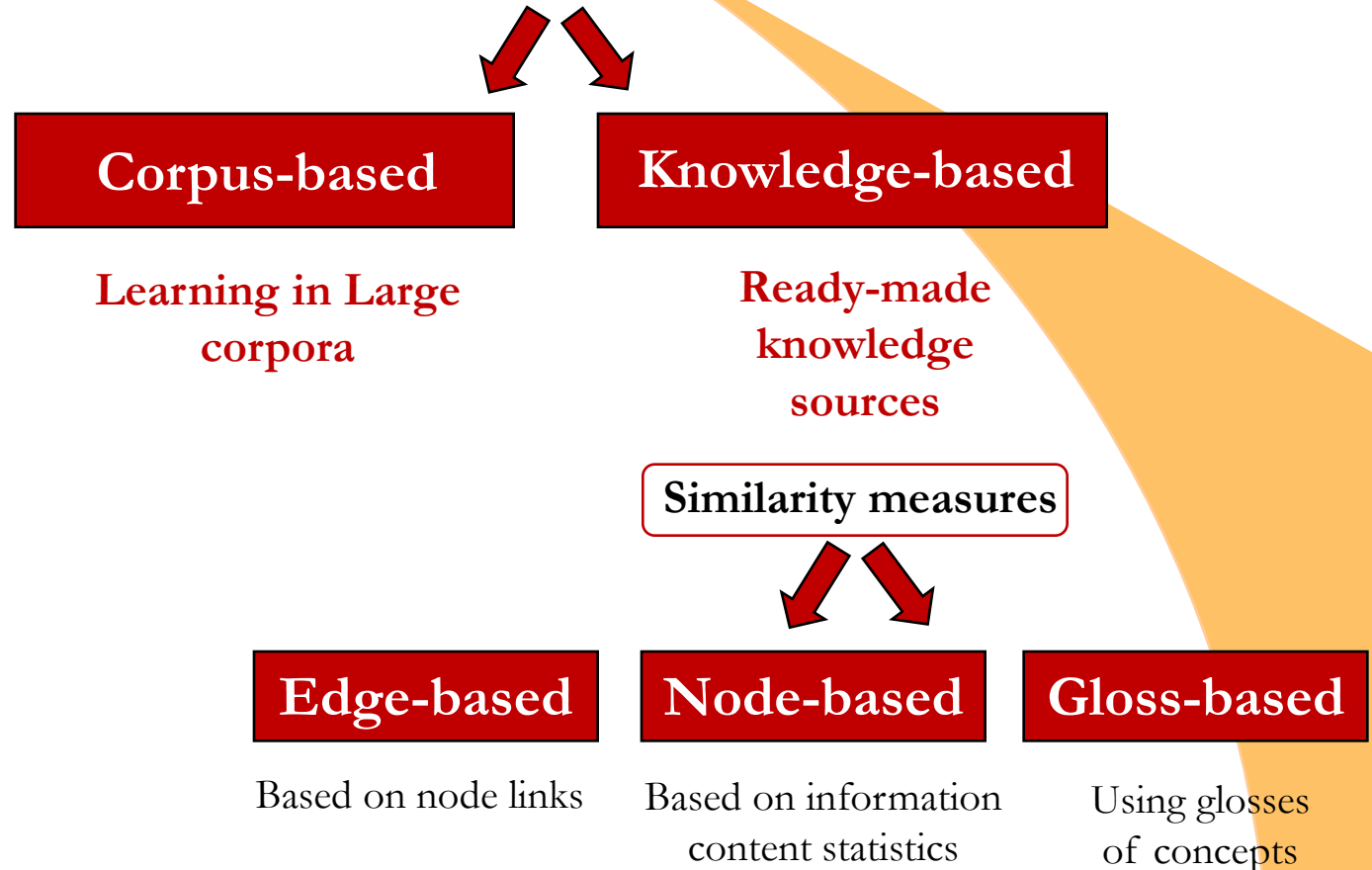
Corpus-based

Knowledge-based

Applications

Conclusion

Semantic Analysis



Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

- ◆ **Edge-based semantic similarity**
 - ◆ **Pro:** Intuitive
 - ◆ **Con:** Depends on varying link distances
- ◆ **Node-based semantic similarity**
 - ◆ **Pro:** Gets round varying link distances
 - ◆ **Con:** Requires corpus-based statistics
- ◆ **Gloss-based semantic similarity**
 - ◆ **Pro:** Makes use of additional information in glosses
 - ◆ **Con:** Limited use of semantic relations between concepts

Empirical results usually show that node-based and integrated methods are generally better predictors of semantic similarity

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

◆ Applications and active research areas

- ◆ Information retrieval
 - **Document and query analysis**
- ◆ Information extraction
 - **Text categorization**
- ◆ Content analysis
 - **Blog classification and social networks**
- ◆ Machine translation
 - **Live translation**
- ◆ Lexicography
 - **Dictionary creation**

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

◆ Open problems and future directions

◆ Performance

◆ Generalization capability

- ▢ Dimension of reference knowledge base
- ▢ Granularity of senses

**Human language is subject to
change and interpretation**

- **Automatic Creation of new word senses**
- **Automatic Merging of word senses**
 - Association rules
 - Lexico-syntactic analysis and parsing
 - Statistical models
 - Learning techniques

Overview

Introduction

Corpus-based

Knowledge-
based

Applications

Conclusion

◆ Open problems and future directions

- ◆ Automatic knowledge base construction
 - For domain specific applications
- ◆ Integration between knowledge bases
 - Between various domain specific sources
- ◆ Domain specific semantic similarity measures
 - Unconventional: using Wikipedia, google, flickr, etc.



Thank you!

Questions ...



Personal Assignments

- ◆ **Manipulating the WorldNet** lexical library and related tools, available at <http://wordnet.princeton.edu/>
- ◆ **Manipulating RDF parsers:**
 - ◆ Java-based Jena RDF, available at <http://jena.apache.org/>
 - ◆ C-based Raptor RDF, available at <http://librdf.org/raptor/>
- ◆ **Manipulating the Protégé** ontology editor tool, available at <http://protege.stanford.edu/>

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- ◆ B. Sathiya, T. V. Geetha: A review on semantic similarity measures for ontology. *J. Intelligent Fuzzy Systems*, 36(4): 3045-3059 (2019)
- ◆ Alexander Budanitsky and Graeme Hirst: Evaluating WordNet-based Measures of Lexical Semantic Relatedness. *Computational Linguistics* 32(1): 13-47, 2006.
- ◆ Roberto Navigli: Word sense disambiguation: A survey. *ACM Computing Surveys* 41(2), 2009.
- ◆ Various other scientific papers and publications...